

Performance Comparison of CNN From Scratch and MobileNetV2 Transfer Learning for Flower Image Classification

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Submitted :

Revision :

Accepted:

Published:

Abstrak–Background: Flower image classification has become an important application of deep learning in agriculture, education, and biodiversity identification. Developing an accurate and efficient classification model remains a significant challenge. **Objective:** This study aims to compare the performance of a CNN From Scratch model and a MobileNetV2 Transfer Learning model with fine-tuning for flower image classification. **Methods:** A quantitative experimental approach was employed using a dataset consisting of seven flower classes: *daisy*, *dandelion*, *lily*, *orchid*, *rose*, *sunflower*, and *tulip*. The research involved image preprocessing, model training, validation, testing, and performance evaluation using accuracy metrics and confusion matrices. **Results:** The CNN From Scratch model achieved a training accuracy of **95.85%**, validation accuracy of **93.76%**, and testing accuracy of **94.19%**. Meanwhile, MobileNetV2 Transfer Learning with fine-tuning obtained **99.92%** training accuracy, **99.39%** validation accuracy, and **99.39%** testing accuracy, with fewer classification errors across all classes. **Conclusion:** The experimental results demonstrate that MobileNetV2 Transfer Learning outperforms CNN From Scratch in terms of accuracy and generalization capability, making it a more effective approach for flower image classification tasks.

Keywords: *Deep Learning; Convolutional Neural Network; MobileNetV2; Transfer Learning; Flower Image Classification; Computer Vision;*

1. INTRODUCTION

The rapid advancement of Artificial Intelligence (AI) has significantly transformed the field of computer vision, particularly in image classification. Computer vision enables computers to interpret and analyze visual information similarly to human perception, making it applicable in various domains such as agriculture, healthcare, environmental monitoring, and industrial automation [1]. Among numerous computer vision tasks, image classification has attracted considerable attention due to its ability to automatically identify object categories from digital images. The increasing availability of large-scale image datasets and improvements in computational resources have accelerated the adoption of deep learning techniques for solving complex image recognition problems [2].

Convolutional Neural Network (CNN) has become one of the most successful deep learning architectures for image classification because of its capability to automatically extract hierarchical image features without requiring handcrafted feature engineering. Unlike conventional machine learning methods, CNN performs feature extraction and classification simultaneously through multiple convolutional, pooling, and fully connected layers. This capability allows CNN to learn complex visual representations from raw image data, resulting in superior performance across numerous computer vision applications [3]. Consequently, CNN has been widely adopted in applications such as medical image diagnosis, traffic sign recognition, plant disease identification, and flower species classification.

Flower image classification represents a challenging computer vision problem because many flower species share similar visual characteristics, including petal color, texture, shape, and overall appearance. Variations in lighting conditions, camera viewpoints, object scales, and background complexity further increase the difficulty of distinguishing flower species accurately. In practical applications, an automatic flower classification system can assist botanical researchers, educational institutions, agricultural industries, and biodiversity conservation programs by providing rapid and accurate species identification [4]. Therefore, developing an accurate and computationally efficient classification model remains an active research topic.

Several deep learning approaches have been proposed to improve flower image classification performance. Conventional CNN models are commonly trained from scratch using randomly initialized parameters. Although this approach enables the network to learn dataset-specific features, it generally requires a large amount of labeled training data, extensive computational resources, and longer training time. Furthermore, CNN models trained from scratch are more susceptible to overfitting when the available dataset is relatively limited [5]. Consequently, researchers have increasingly adopted transfer learning techniques to overcome these limitations.

Transfer learning has become one of the most effective approaches in deep learning by utilizing knowledge learned from large-scale benchmark datasets such as ImageNet. Instead of training an entire network from random initialization, pretrained models reuse previously learned feature representations and adapt them to a new classification task through feature extraction and fine-tuning. This strategy significantly reduces training time while improving model convergence and classification performance, particularly when the available dataset is relatively small [6].

Among various pretrained CNN architectures, MobileNetV2 has gained considerable popularity because it is specifically designed for lightweight image classification. MobileNetV2 employs depthwise separable convolution and inverted residual blocks with linear bottlenecks, allowing the network to achieve high classification accuracy while maintaining low computational complexity and memory consumption [7]. These characteristics make MobileNetV2 suitable for implementation on mobile devices, embedded systems, and Internet of Things (IoT) platforms where computational resources are limited.

Several recent studies have demonstrated the effectiveness of transfer learning for image classification tasks. Rahman *et al.* proposed a flower image classification system using MobileNetV2 and transfer learning, reporting substantial improvements in classification accuracy compared with conventional CNN models trained from scratch [8]. Similarly, Simangunsong *et al.* evaluated transfer learning techniques for flower classification and concluded that pretrained CNN architectures provide better feature representation and faster convergence than conventional CNN training methods [9]. Another comparative study conducted by Yulisara *et al.* compared several CNN architectures and optimization algorithms for flower image classification, demonstrating that MobileNetV2 consistently achieved competitive performance while requiring relatively lower computational cost than deeper architectures [10].

Although previous studies have successfully demonstrated the advantages of transfer learning, several limitations remain. Most existing research focuses primarily on evaluating pretrained architectures without directly comparing them against a CNN model developed entirely from scratch under identical experimental settings. Furthermore, many studies emphasize overall accuracy while providing limited discussion regarding confusion matrices, class-wise performance, and generalization capability across multiple flower categories. Consequently, the actual performance difference between CNN From Scratch and MobileNetV2 Transfer Learning using identical datasets, preprocessing techniques, and evaluation metrics remains insufficiently investigated.

Although previous studies have demonstrated that transfer learning generally improves image classification performance, several research gaps remain. First, most studies focus only on evaluating pretrained architectures without conducting a direct comparison with a CNN model developed entirely from scratch under identical experimental conditions [11]. Consequently, the actual performance improvement provided by transfer learning cannot be objectively quantified because differences in datasets, preprocessing techniques, and training configurations may influence the experimental results.

Second, several previous studies mainly report overall classification accuracy while providing limited discussion regarding class-wise performance using confusion matrices and classification reports [12]. In image classification problems involving visually similar flower species, relying solely on accuracy may hide classification errors occurring in specific classes. Therefore, a more comprehensive evaluation involving precision, recall, F1-score, and confusion matrix analysis is required to better understand the strengths and weaknesses of each model.

Third, although MobileNetV2 has been widely recognized as an efficient lightweight convolutional neural network, only a limited number of studies have investigated its performance using fine-tuning techniques on flower image datasets containing multiple species with similar visual characteristics [13]. Fine-tuning enables pretrained feature representations to adapt to domain-specific image characteristics and may significantly improve model generalization capability.

Furthermore, previous comparative studies generally emphasize model accuracy while paying less attention to computational efficiency and training convergence [14]. These aspects are important because lightweight architectures are increasingly deployed on mobile devices and embedded systems with limited computational resources. Consequently, evaluating both classification performance and training behavior provides a more comprehensive understanding of model effectiveness.

Based on these research gaps, this study proposes a comprehensive comparison between **CNN From Scratch** and **MobileNetV2 Transfer Learning with Fine-Tuning** using identical datasets, preprocessing procedures, training parameters, and evaluation metrics. The comparison includes training accuracy, validation accuracy, testing accuracy, loss curves, confusion matrices, and classification reports to ensure an objective evaluation of both architectures [15].

The main objective of this research is to analyze and compare the performance of CNN From Scratch and MobileNetV2 Transfer Learning for classifying seven flower categories, namely **daisy, dandelion, lily, orchid, rose, sunflower, and tulip**. In addition, this study aims to identify the advantages of transfer learning in improving classification accuracy, reducing classification errors, and enhancing model generalization compared with conventional CNN training [16].

The contribution of this research lies in providing an experimental comparison conducted under identical conditions, enabling a fair evaluation of both deep learning approaches. The findings are expected to provide

practical guidance for researchers and developers in selecting an appropriate CNN architecture for flower image classification. Furthermore, the proposed approach may serve as a reference for developing lightweight computer vision applications on mobile and embedded devices that require high classification accuracy with relatively low computational cost [17].

2. RESEARCH METHOD

2.1 Research Design

This study employed a quantitative experimental approach to compare the performance of two deep learning models for flower image classification: **CNN From Scratch** and **MobileNetV2 Transfer Learning with Fine-Tuning**. Both models were trained and evaluated using the same dataset, preprocessing techniques, hyperparameter settings, and evaluation metrics to ensure a fair comparison. The overall research workflow consisted of dataset preparation, image preprocessing, model development, model training, performance evaluation, and comparative analysis.

The objective of this experiment was to identify which model provides better classification accuracy and generalization capability for recognizing different flower species. Figure 1 illustrates the overall research workflow.

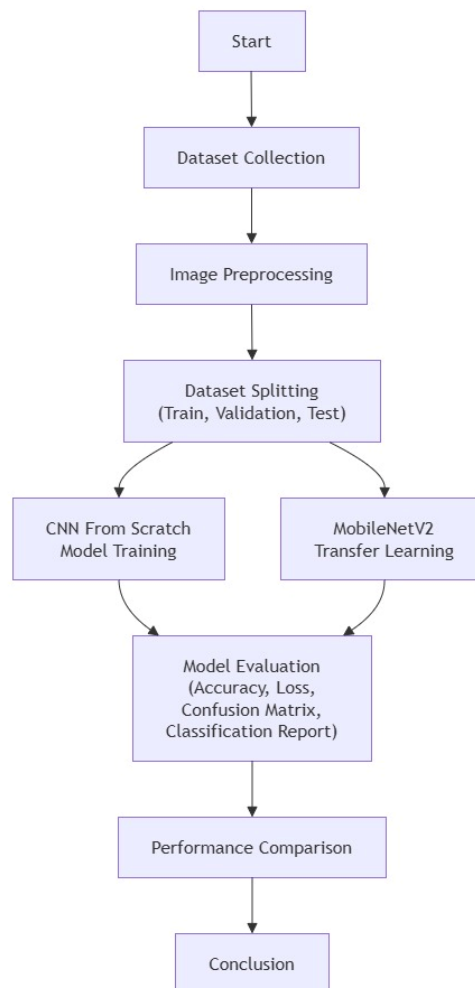


Figure 1. Research Stages

2.2 Dataset Preparation

The dataset used in this research consists of seven flower categories, namely **daisy, dandelion, lily, orchid, rose, sunflower, and tulip**. Each image represents a single flower object with variations in illumination, background, object orientation, and scale. These variations increase the complexity of the classification task and provide a more realistic evaluation scenario.

Before training, the dataset was organized into three subsets consisting of training, validation, and testing data. This data partition allows the model to learn from the training set while preventing overfitting through validation

and measuring its generalization performance using previously unseen testing images. The same dataset partition was used for both CNN architectures to maintain experimental consistency.

2.3 Image Preprocessing

Image preprocessing was performed to standardize all input images before they were processed by the neural networks. Each image was resized into a fixed input dimension compatible with both CNN From Scratch and MobileNetV2. Pixel values were normalized into floating-point values ranging between 0 and 1 to accelerate model convergence during training.

Data preprocessing also included image augmentation to improve model robustness against variations in image appearance. Augmentation techniques such as random rotation, horizontal flipping, zooming, and shifting were applied to increase data diversity without changing the original class labels. These preprocessing techniques help reduce overfitting and improve the model's ability to recognize unseen images.

2.4 CNN From Scratch Architecture

The first model was developed using a Convolutional Neural Network constructed entirely from scratch. The network consisted of several convolutional layers followed by Rectified Linear Unit (ReLU) activation functions and max-pooling layers to extract hierarchical image features. The extracted features were flattened before passing through fully connected layers for classification.

The output layer employed the Softmax activation function to classify images into seven flower categories. Model optimization was performed using the Adam optimizer with categorical cross-entropy as the loss function. During training, model performance was monitored using training accuracy, validation accuracy, training loss, and validation loss.

2.5 MobileNetV2 Transfer Learning

The second model utilized the MobileNetV2 architecture pretrained on the ImageNet dataset. Instead of training the network from random initialization, pretrained weights were used as the feature extractor. Fine-tuning was then performed by unfreezing selected upper layers so that the network could adapt its learned features to the flower classification dataset.

The classification head of MobileNetV2 was modified by adding a Global Average Pooling layer, fully connected layer, dropout layer, and Softmax output layer consisting of seven neurons corresponding to the seven flower classes. Transfer learning significantly reduced training time while improving classification performance because the pretrained model had already learned rich visual representations from millions of natural images.

2.6 Model Training

Both CNN models were trained using identical experimental settings to ensure objective comparison. During training, the optimizer updated network weights by minimizing categorical cross-entropy loss through backpropagation. Validation data were evaluated after each training epoch to monitor model learning and detect potential overfitting.

Training performance was recorded using two primary indicators:

- Training accuracy
- Validation accuracy

Additionally, training and validation loss curves were analyzed to evaluate convergence behavior throughout the learning process.

2.7 Performance Evaluation

The performance of both models was evaluated using the independent testing dataset. Several evaluation metrics were employed to comprehensively measure classification performance, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

Accuracy was used as the primary indicator of overall classification performance, while precision, recall, and F1-score provided class-wise evaluation. Furthermore, confusion matrices were analyzed to identify correctly classified images as well as misclassified flower categories. These evaluation metrics provide a more comprehensive assessment than overall accuracy alone because they reveal the strengths and weaknesses of each model for individual flower classes.

2.8 Comparative Analysis

The final stage compared CNN From Scratch and MobileNetV2 Transfer Learning based on several performance indicators, including training accuracy, validation accuracy, testing accuracy, loss curves, confusion matrices, and classification reports. The comparison focused on identifying the most effective architecture for flower image classification while considering both classification performance and model generalization capability.

The model achieving higher testing accuracy together with lower classification errors and more stable training behavior was considered the better approach for automatic flower image classification. The findings from this comparative analysis serve as the basis for determining the advantages and limitations of both CNN architectures in practical computer vision applications.

3. RESULTS AND DISCUSSION

This section presents and discusses the experimental results obtained from the implementation of two deep learning approaches for flower image classification, namely CNN From Scratch and MobileNetV2 Transfer Learning with Fine-Tuning. The evaluation was conducted using the same dataset and data partition to ensure a fair comparison between the two models. The analysis includes dataset distribution, training and validation performance, testing performance, confusion matrices, classification reports, and overall model comparison. The results are discussed to determine the effectiveness and generalization capability of each model in classifying seven flower categories: *daisy*, *dandelion*, *lily*, *orchid*, *rose*, *sunflower*, and *tulip*.

3.1 Dataset Distribution

The dataset used in this study consists of seven flower classes with different visual characteristics. The classes include *daisy*, *dandelion*, *lily*, *orchid*, *rose*, *sunflower*, and *tulip*. Before model training, the dataset was divided into training, validation, and testing subsets. The purpose of this division was to separate the data used for parameter learning, model monitoring, and final performance evaluation.

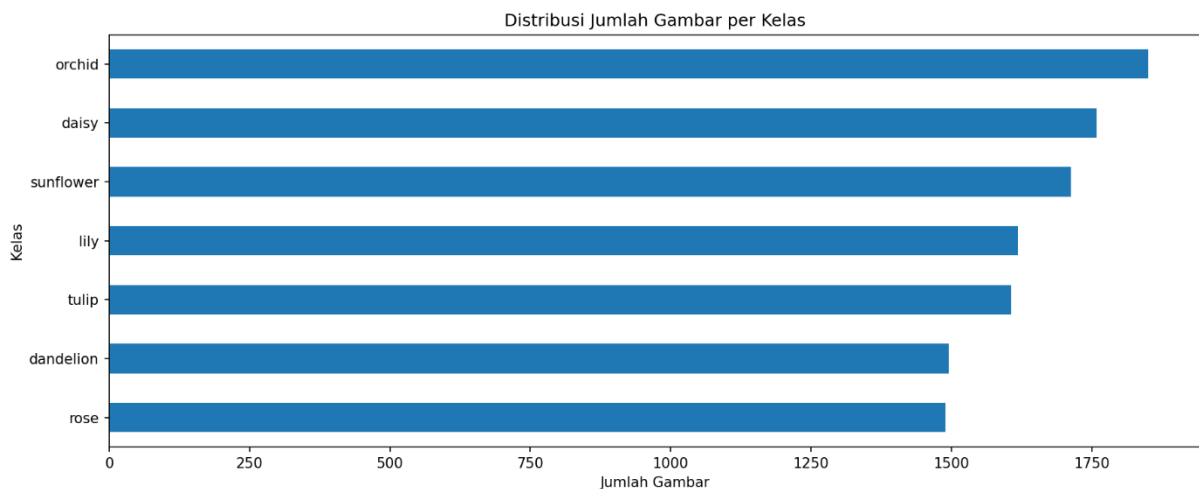


Figure 2. Distribution of Flower Image Classes.

Figure 2 presents the distribution of images among the seven flower classes. The dataset contains variations in the number of images for each class; however, the distribution remains sufficiently representative for multiclass classification. The testing dataset consists of **1,154 images**, including **176 daisy images**, **149 dandelion images**, **162 lily images**, **185 orchid images**, **149 rose images**, **172 sunflower images**, and **161 tulip images**.

The *orchid* class contains the largest number of testing samples with 185 images, whereas the *dandelion* and *rose* classes each contain 149 testing images. Although the number of samples differs slightly between classes, the difference is not excessively large. Therefore, the evaluation results can still provide a reasonable representation of model performance across all seven classes. In addition to overall accuracy, macro-average and weighted-average metrics were used to evaluate whether the models performed consistently across the different classes.

The use of the same training, validation, and testing partitions for both models is important because it ensures that the performance comparison is conducted under equivalent experimental conditions. Therefore, differences in classification performance can be primarily attributed to differences in model architecture and learning strategy rather than differences in the evaluated data.

3.2 Performance of CNN From Scratch

The first experiment evaluated the CNN From Scratch model. Unlike pretrained architectures, this model learned all feature representations directly from the flower image dataset. The training process was monitored using accuracy and loss values on both the training and validation datasets.

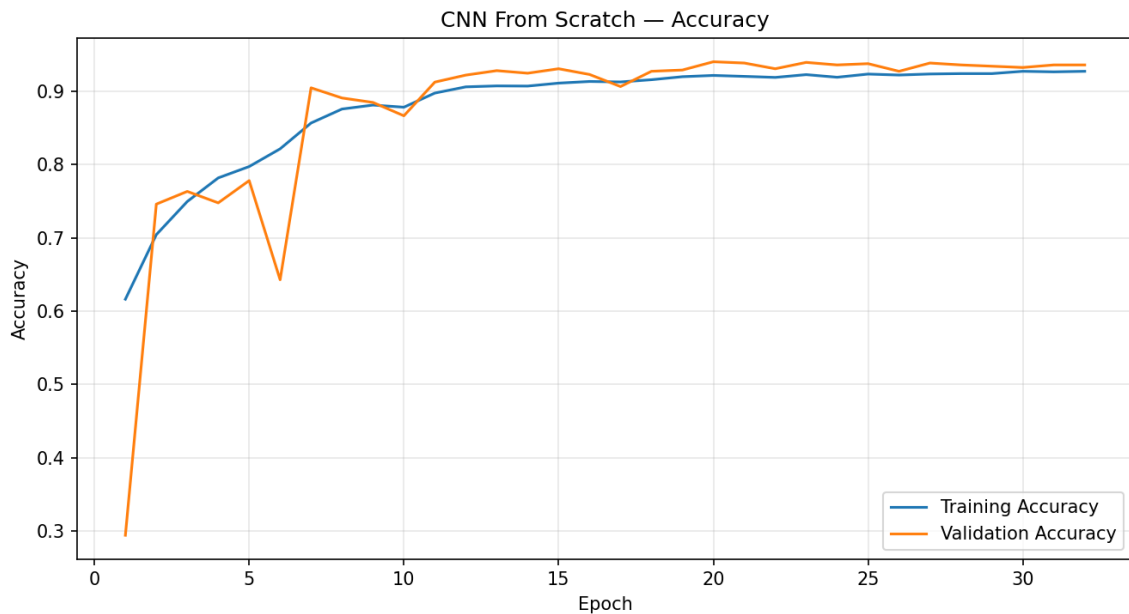


Figure 3. Training and Validation Accuracy of CNN From Scratch

Figure 3 shows the development of training and validation accuracy during the CNN From Scratch training process. The model achieved a final **training accuracy of 95.85%** and a **validation accuracy of 93.76%**. The relatively small difference between training and validation accuracy indicates that the model was capable of learning relevant visual patterns without experiencing severe overfitting.

The training accuracy was approximately 2.09 percentage points higher than the validation accuracy. This difference is considered relatively small for an image classification task because the model is naturally expected to perform slightly better on data used during the optimization process. The validation performance remained close to the training performance, indicating that the learned features could be generalized to images that were not directly used to update the model parameters.

The testing process produced an accuracy of **94.19%**, which was slightly higher than the validation accuracy. This result supports the conclusion that the CNN From Scratch model achieved a **good fit**. The model did not exhibit a substantial decrease in performance when evaluated on unseen testing data. Therefore, although it was trained without pretrained weights, the model successfully learned meaningful visual representations from the seven flower classes.

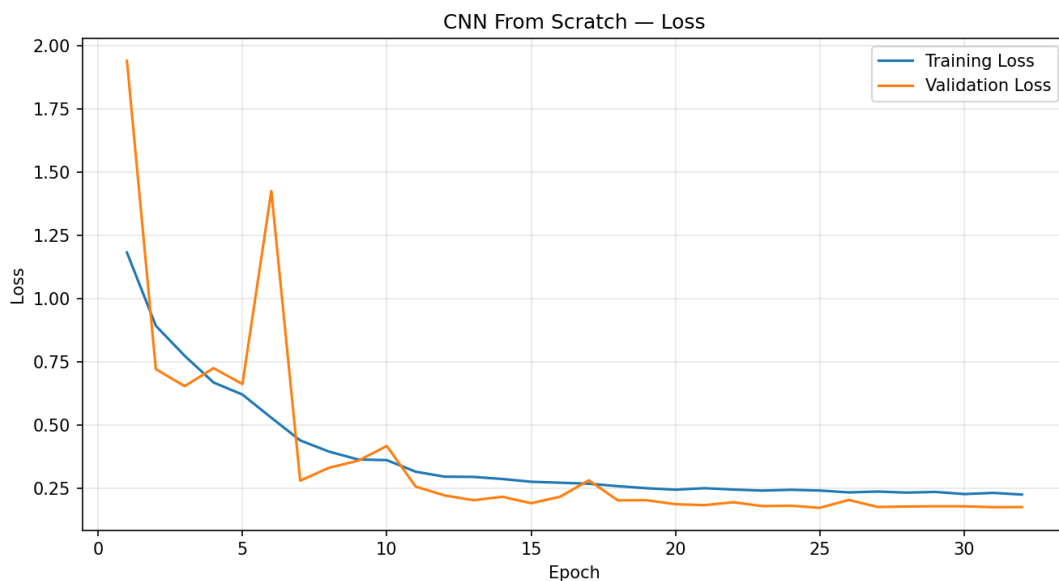


Figure 4. Training and Validation Loss of CNN From Scratch

The loss curves shown in Figure 4 provide additional information regarding the model's learning behavior. The final training loss was **0.132987**, while the validation loss was **0.174532**. The testing loss was **0.188053**. The relatively small difference among these values indicates that the model maintained stable generalization performance.

The training loss was lower than the validation and testing losses, which is an expected pattern because the model parameters were directly optimized using the training dataset. However, the difference was not large enough to indicate severe overfitting. The combination of high accuracy and relatively stable loss values demonstrates that the CNN From Scratch architecture successfully converged during training.

To provide a more detailed evaluation, the predictions generated by the model were analyzed using a confusion matrix.

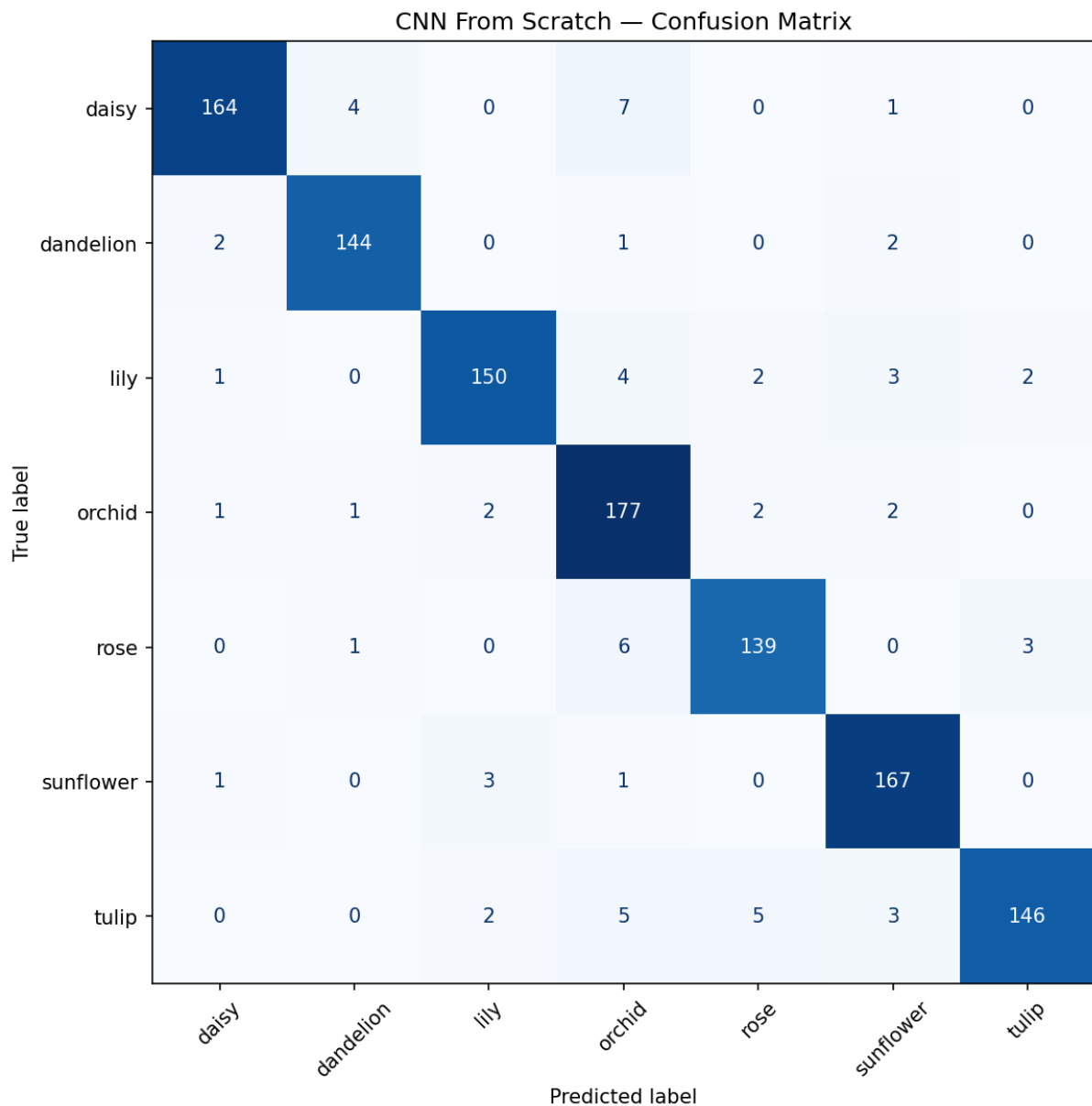


Figure 5. Confusion Matrix of CNN From Scratch.

The confusion matrix in Figure 5 demonstrates that most testing images were correctly classified into their respective flower categories. However, several misclassifications remained, particularly among flower classes with similar visual characteristics.

The classification report shows that the *daisy* class achieved a precision of **97.04%**, recall of **93.18%**, and F1-score of **95.07%**. This result indicates that predictions classified as *daisy* were generally accurate, although several actual daisy images were assigned to other classes.

The *dandelion* class achieved a precision of **96.00%**, recall of **96.64%**, and F1-score of **96.32%**, making it one of the strongest classes in the CNN From Scratch model. The balanced precision and recall values indicate that the model was consistently capable of recognizing dandelion images.

For the *lily* class, the model obtained **95.54% precision**, **92.59% recall**, and an F1-score of **94.04%**. The lower recall compared with precision indicates that some actual lily images were incorrectly predicted as other flower categories.

The *orchid* class presented an interesting result. It achieved the lowest precision among all classes at **88.06%**, but its recall reached **95.68%**, resulting in an F1-score of **91.71%**. This means that the model successfully detected most actual orchid images but also incorrectly classified several images from other categories as orchids. Therefore, the CNN From Scratch model tended to produce more false-positive predictions for the orchid class.

The *rose* class achieved a precision of **93.92%**, recall of **93.29%**, and F1-score of **93.60%**. Meanwhile, the *sunflower* class achieved a precision of **93.82%**, recall of **97.09%**, and F1-score of **95.43%**. The high recall for sunflower indicates that the model successfully identified most sunflower images in the testing dataset.

The *tulip* class achieved a high precision of **96.69%**, but its recall was lower at **90.68%**, producing an F1-score of **93.59%**. This result suggests that when the model predicted an image as a tulip, the prediction was usually correct. However, several actual tulip images were classified into other flower categories.

Overall, the CNN From Scratch model obtained a **macro-average precision of 94.44%**, **macro-average recall of 94.17%**, and **macro-average F1-score of 94.25%**. Its weighted-average precision, recall, and F1-score were **94.32%**, **94.19%**, and **94.21%**, respectively. These results demonstrate that the model provided relatively balanced classification performance across the seven flower categories.

3.3 Performance of MobileNetV2 Transfer Learning with Fine-Tuning

The second experiment employed MobileNetV2 using a transfer learning and fine-tuning strategy. Unlike the CNN From Scratch model, MobileNetV2 utilized pretrained feature representations. These pretrained features were subsequently adapted to the flower classification task through fine-tuning.

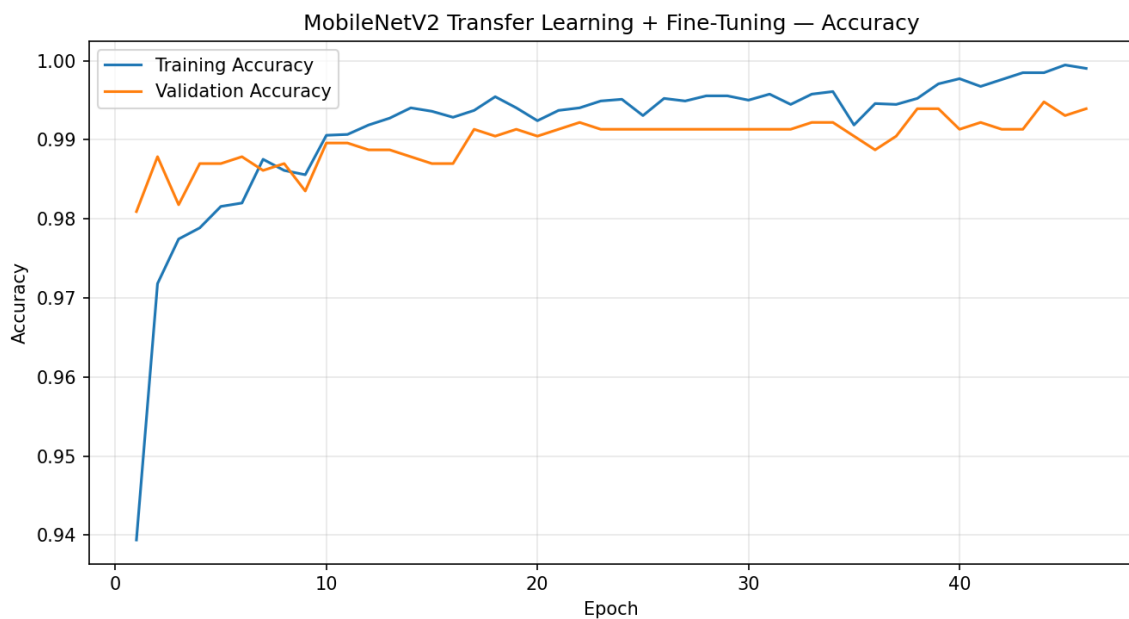


Figure 6. Training and Validation Accuracy of MobileNetV2.

The MobileNetV2 model achieved a **training accuracy of 99.92%** and a **validation accuracy of 99.39%**. The difference between these values was only approximately 0.53 percentage points. This small performance gap indicates excellent generalization capability and suggests that the model did not experience significant overfitting. The testing accuracy was **99.39%**, which was almost identical to the validation accuracy. This consistency is particularly important because it demonstrates that the performance observed during model development was maintained when the model was evaluated using independent testing images.

Compared with the CNN From Scratch model, MobileNetV2 achieved considerably higher accuracy on all three dataset partitions. This improvement demonstrates the effectiveness of transfer learning. The pretrained MobileNetV2 model had previously learned general visual representations, such as edges, shapes, textures, and more complex patterns. Fine-tuning allowed these features to be adapted to the specific characteristics of the seven flower classes.

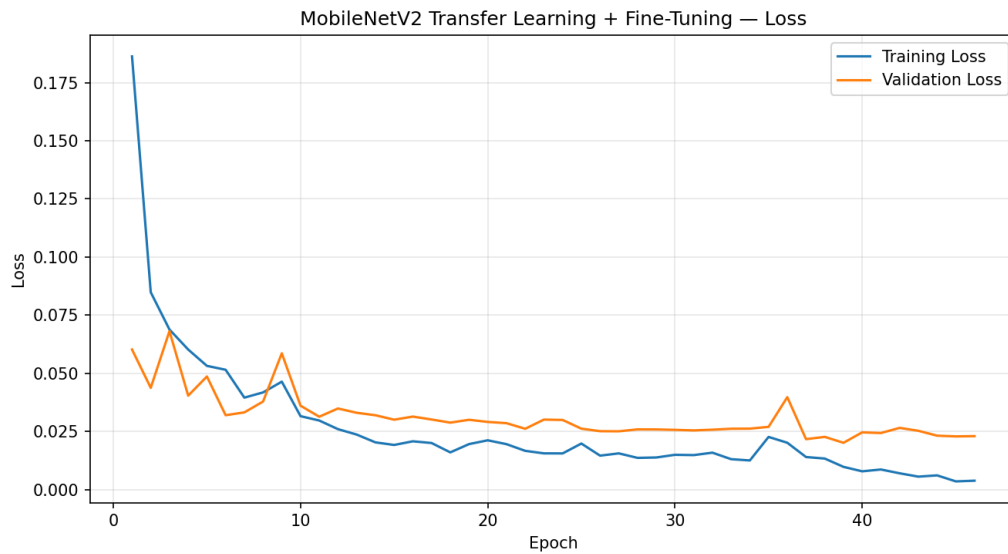


Figure 7. Training and Validation Loss of MobileNetV2.

The loss results further demonstrate the strong performance of MobileNetV2. The final training loss was only **0.003309**, while the validation loss was **0.020112** and the testing loss was **0.028295**. These values were substantially lower than those produced by the CNN From Scratch model.

Although the validation and testing losses were higher than the training loss, their absolute values remained very low. Combined with validation and testing accuracies of approximately 99.39%, these results indicate that the model learned highly discriminative representations for the seven flower categories.

The consistency between validation and testing performance also provides strong evidence that the model generalized effectively. The model was therefore categorized as **GOOD FIT**, similar to the CNN From Scratch model, but with significantly better overall performance.

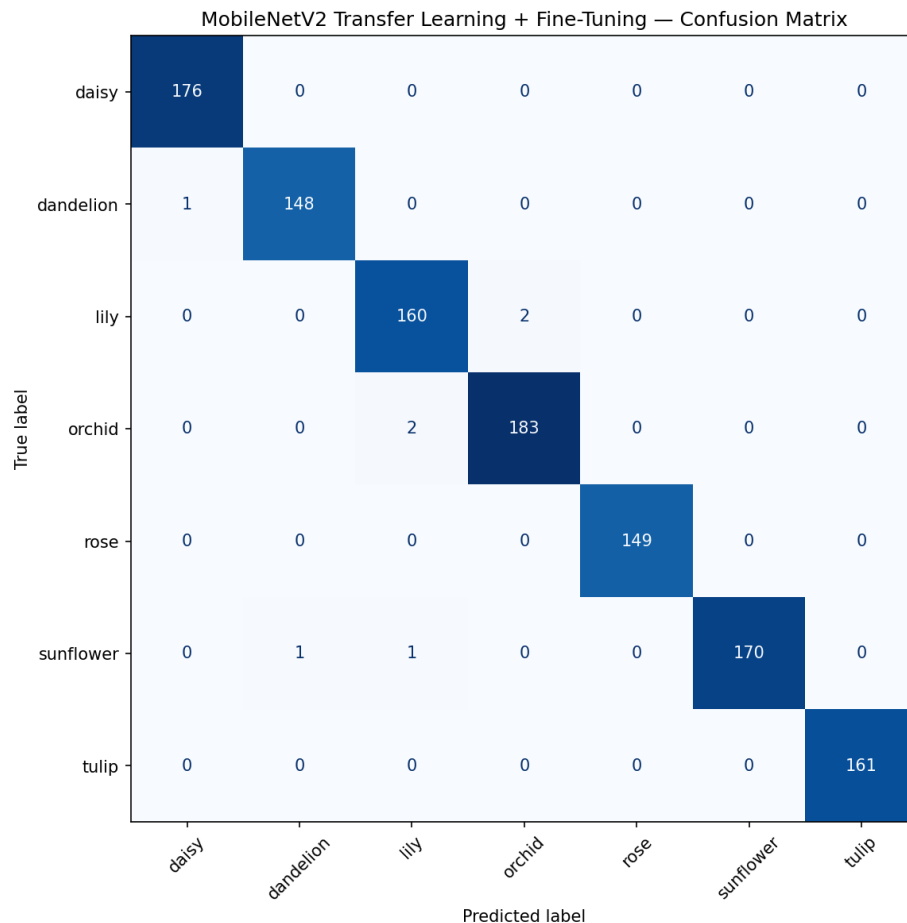


Figure 8. Confusion Matrix of MobileNetV2 Transfer Learning with Fine-Tuning.

The MobileNetV2 confusion matrix demonstrates a substantial reduction in classification errors compared with CNN From Scratch. Most predictions are concentrated along the main diagonal of the matrix, indicating that almost all testing images were assigned to their correct categories.

The *daisy* class achieved **99.44% precision, 100% recall**, and an F1-score of **99.72%**. A recall of 100% indicates that all 176 daisy images in the testing dataset were successfully recognized.

The *dandelion* class obtained **99.33% precision, 99.33% recall**, and **99.33% F1-score**. This balanced performance demonstrates that only a very small number of classification errors occurred for this category.

For the *lily* class, MobileNetV2 achieved **98.16% precision, 98.77% recall**, and an F1-score of **98.46%**. Although this was the lowest F1-score among the seven classes, the result remained substantially higher than the corresponding performance of the CNN From Scratch model.

The *orchid* class achieved **98.92% precision, 98.92% recall**, and **98.92% F1-score**. This represents a major improvement over the CNN From Scratch model, particularly in precision, which increased from 88.06% to 98.92%. This improvement indicates that the pretrained features of MobileNetV2 were more effective in distinguishing orchids from visually similar flower categories.

The *rose* class achieved perfect performance with **100% precision, 100% recall**, and **100% F1-score**. This means that all rose images were correctly identified without false-positive or false-negative predictions. Similarly, the *tulip* class also achieved **100% precision, recall, and F1-score**.

The *sunflower* class achieved **100% precision, 98.84% recall**, and an F1-score of **99.42%**. The perfect precision indicates that every image predicted as a sunflower was correct, while the slightly lower recall indicates that only a very small number of actual sunflower images were assigned to another category.

Overall, MobileNetV2 obtained a **macro-average precision of 99.41%, macro-average recall of 99.41%**, and **macro-average F1-score of 99.41%**. The weighted-average precision, recall, and F1-score were **99.40%, 99.39%, and 99.39%**, respectively. The similarity between macro-average and weighted-average results indicates that the model performed consistently across all seven classes.

3.4 Comparison of CNN From Scratch and MobileNetV2

A direct comparison between the two models was conducted using training, validation, and testing performance.

Table 1. Performance Comparison of the Proposed Models.

Model	Training Accuracy	Validation Accuracy	Testing Accuracy	Testing Loss	Fit Status
CNN From Scratch	95.85%	93.76%	94.19%	0.1881	Good Fit
MobileNetV2 Transfer Learning + Fine-Tuning	99.92%	99.39%	99.39%	0.0283	Good Fit

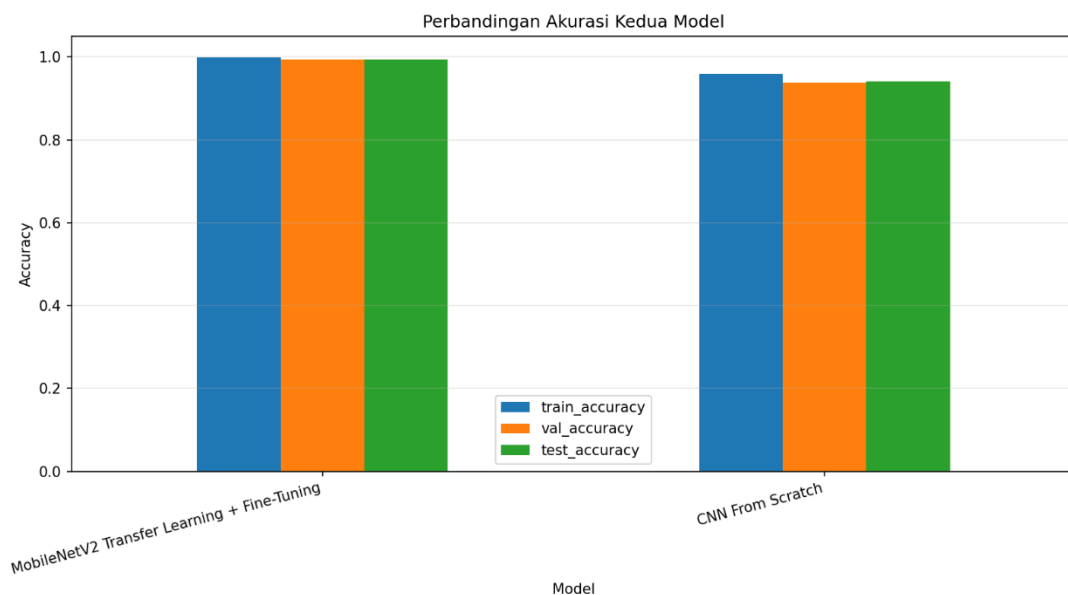


Figure 9. Accuracy Comparison Between CNN From Scratch and MobileNetV2.

As shown in Table 1 and Figure 9, MobileNetV2 outperformed CNN From Scratch in all accuracy measurements. The difference in testing accuracy was approximately **5.20 percentage points**, increasing from **94.19%** for CNN From Scratch to **99.39%** for MobileNetV2.

Although a difference of 5.20 percentage points may initially appear moderate, its practical impact is substantial. On a testing dataset containing 1,154 images, the CNN From Scratch accuracy corresponds to approximately 67 incorrectly classified images, whereas the MobileNetV2 accuracy corresponds to only approximately 7 incorrect predictions. Therefore, MobileNetV2 reduced the number of classification errors by approximately 60 images in the testing dataset.

The improvement can be attributed primarily to the transfer learning strategy. CNN From Scratch must learn all visual representations directly from the available flower images. In contrast, MobileNetV2 begins with pretrained feature representations that already encode general visual patterns. Fine-tuning then adapts these representations to the specific flower classification task.

The results also show that both models were categorized as **GOOD FIT**. Therefore, the CNN From Scratch model cannot be considered unsuccessful. It achieved high testing accuracy and balanced class-wise performance. However, MobileNetV2 demonstrated a significantly stronger ability to distinguish visually similar flower categories and produced fewer classification errors.

Another important difference is visible in the class-wise metrics. The CNN From Scratch model showed greater variation in precision and recall across classes. For example, orchid precision was only 88.06%, while sunflower recall reached 97.09%. In contrast, MobileNetV2 maintained precision and recall values above 98% for all seven classes. This consistency demonstrates that the performance advantage of MobileNetV2 was not limited to a particular class but was distributed across the entire classification task.

These findings support the research objective of evaluating the effectiveness of transfer learning compared with a model trained entirely from scratch. Under the same classification task, MobileNetV2 Transfer Learning with Fine-Tuning achieved higher accuracy, lower loss, fewer misclassifications, and more consistent per-class performance. Therefore, MobileNetV2 was selected as the best-performing model in this study.

3.5 TensorFlow Lite Inference Evaluation.

After determining MobileNetV2 as the best-performing model, an inference evaluation was also prepared using the TensorFlow Lite format. TensorFlow Lite is commonly used to convert deep learning models into a format suitable for deployment on mobile and resource-constrained devices.

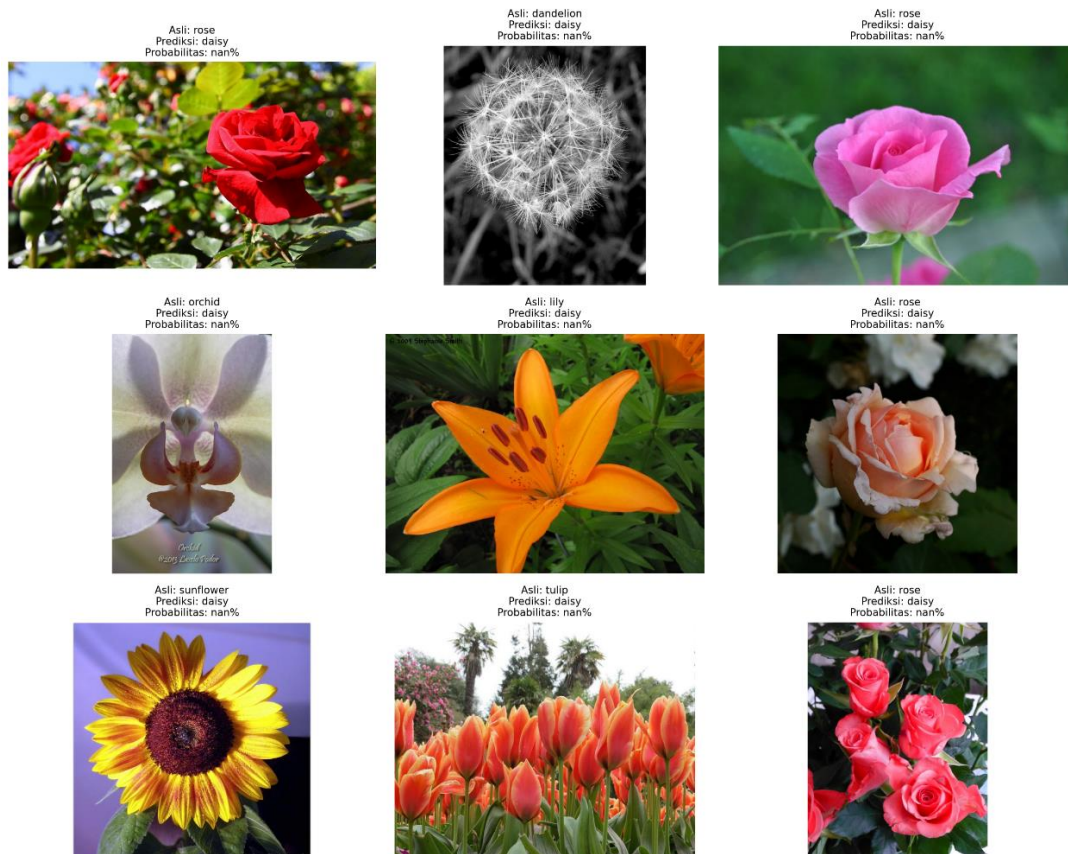


Figure 10. Sample Results of TensorFlow Lite Model Inference.

However, based on the provided TensorFlow Lite inference result file, the sample inference output shows that the prediction probability values were not successfully recorded and several sample images were predicted as *daisy*. Therefore, this specific TensorFlow Lite inference result should **not yet be interpreted as evidence of successful deployment performance**.

This result indicates that the TensorFlow Lite inference pipeline requires further verification. The problem may be related to input preprocessing, input tensor data type, normalization, output interpretation, or class index mapping between the original model and the converted TensorFlow Lite model. Because the original MobileNetV2 model achieved a testing accuracy of 99.39%, a major performance degradation after conversion would indicate that the deployment pipeline should be checked rather than immediately concluding that the trained model itself performs poorly.

Therefore, the primary experimental conclusion of this study is based on the original evaluated CNN From Scratch and MobileNetV2 models. The TensorFlow Lite stage should be considered an additional implementation experiment that requires further validation. Future implementation should ensure that the preprocessing procedure used during TensorFlow Lite inference is identical to the preprocessing applied during model training and testing. After the inference pipeline is corrected, the converted model should be reevaluated using the complete testing dataset to determine whether its accuracy remains comparable to the original MobileNetV2 model.

Overall, the experimental findings demonstrate that both models successfully learned the visual characteristics of the seven flower categories. CNN From Scratch achieved strong performance with **94.19% testing accuracy**, demonstrating that a custom convolutional architecture can effectively classify flower images. Nevertheless, MobileNetV2 Transfer Learning with Fine-Tuning achieved the best overall result with **99.39% testing accuracy**, substantially lower testing loss, and highly consistent precision, recall, and F1-score values across all classes. These results confirm that the reuse of pretrained visual features combined with domain-specific fine-tuning provides a significant advantage for multiclass flower image classification.

4. CONCLUSION

This study successfully compared the performance of CNN From Scratch and MobileNetV2 Transfer Learning with Fine-Tuning for the classification of seven flower categories: *daisy*, *dandelion*, *lily*, *orchid*, *rose*, *sunflower*, and *tulip*. The experimental results demonstrate that both models were capable of learning relevant visual features and achieving strong classification performance. The CNN From Scratch model achieved a training accuracy of 95.85%, validation accuracy of 93.76%, and testing accuracy of 94.19%. These results indicate that a custom CNN architecture can effectively perform multiclass flower image classification and maintain good generalization performance on unseen data.

However, MobileNetV2 Transfer Learning with Fine-Tuning demonstrated superior performance, achieving a training accuracy of 99.92%, validation accuracy of 99.39%, and testing accuracy of 99.39%. The model also produced lower classification errors and more consistent precision, recall, and F1-score values across the seven flower classes. Therefore, the results confirm that the use of pretrained visual features combined with fine-tuning provides a significant advantage compared with training a CNN entirely from scratch. Under the experimental conditions used in this study, MobileNetV2 was identified as the best-performing model for flower image classification.

The findings of this study demonstrate that transfer learning can provide an effective approach for developing accurate image classification systems, particularly when compared under the same dataset and evaluation conditions. Future research may extend this study by using a larger and more diverse dataset, comparing MobileNetV2 with other lightweight architectures, and evaluating model efficiency in terms of inference time, model size, and computational requirements. Further development may also include validated deployment on mobile or embedded devices to evaluate the practical performance of the classification model in real-world applications.

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